AWS HACKATHON 2026 // LIVE PROJECT

GuzoMate

ROAD AI ASSISTANT & TRANSIT CO-PILOT

Your AI-powered travel companion for safer journeys across Ethiopia and beyond.
Bridging dense multilingual RAG, real-time spatial intelligence, and 18,833 verified locations.

CORE SYSTEM MODULES

- Places & Spatial Discovery: PostGIS ST_Distance radius queries
- Multilingual AI Assistant: BAAI/bge-m3 dense embeddings + Gemini LLM

TRILINGUAL GROUNDING & SERVICES

- Supported Languages: English • አማርኛ (Amharic) • Afaan Oromoo
- Live Integrations: WeatherAPI, GTFS transit routes, crowd incident radar

THE PROBLEM

Critical Navigation & Travel Blindspots in East Africa

Why standard global mapping applications fall short for everyday commuters and long-distance travelers.

01 // SERVICE DEFICIT

Unmapped Local Services

Global platforms lack verified local mechanics, tire repair shops (ጎማ ጥገና), roadside towing, and informal minibus depots.

02 // HAZARD UNPREDICTABILITY

Dynamic Road Hazards

Escarpment rockfalls, flash flooding, and landslides happen without warning; drivers enter blocked corridors completely blind.

03 // MEDICAL RESPONSE LAG

High-Altitude Emergencies

Critical delays finding oxygen-equipped pharmacies, trauma facilities, or emergency towing in remote highland passes.

04 // LINGUISTIC EXCLUSION

The Language & Script Divide

Over 90% of local travelers navigate in Amharic (Ge'ez) or Afaan Oromo, while traditional mapping tools enforce English-only prompts.

THE SOLUTION

GuzoMate: An Intelligent, Multilingual Travel Co-Pilot

A resilient architecture combining OpenStreetMap discovery, dense multilingual RAG, and community safety.

GEOSPATIAL FOUNDATION

18,833 Verified POIs

- Curated OpenStreetMap locations on Supabase PostgreSQL
- PostGIS ST_Distance high-speed geodesic radius calculations
- Deduplicated within 35m spatial radius for data fidelity
- Dynamic search radius: 3km local → 8km → 18km city-wide

MULTILINGUAL AI

Dense Grounded RAG

- BAAI/bge-m3 1,024-dim multilingual dense embeddings
- Native English, Amharic (አማርኛ), and Afaan Oromoo support
- Gemini 3.5 Flash Lite grounded in strictly delimited evidence
- Zero fabrication constraint with explicit limitation disclosures

SAFETY & TRANSIT

Community Incident Radar

- Geotagged reporting with 500m / 2hr anti-spam duplicate detection
- 3-vote corroboration triggers automatic PENDING → ACTIVE state
- Automated lifecycle expiry (2h traffic to 72h road damage)
- Weather arrival advice matching rain probability to arrival ETA

CORE CAPABILITIES

Three Pillars of Journey Intelligence

Verified functional components powering real-time travel decisions.

01

AI Travel Assistant & Voice Guidance

Natural conversational guidance in English, Amharic (አማርኛ), and Afaan Oromoo.

- Grounded retrieval queries PostGIS points, weather forecasts, and route corridors simultaneously
- Voice transcription (Web Speech API) & speech synthesis for hands-free driver safety
- Sense-of-urgency detection delivers immediate, direct advice during roadside emergencies

02

Nearby Services Discovery (18,833 POIs)

Sub-50ms PostGIS spatial search across 8 verified travel categories.

- Specialized automotive garages, tire repair, and roadside towing facilities
- 24/7 emergency clinics, trauma centers, and oxygen-equipped pharmacies
- Hotels, cultural coffee hubs, gas stations, and intercity bus terminals

03

Road Condition & Incident Radar

Crowdsourced road safety telemetry backed by democratic peer consensus.

- Submit road hazards: accidents, roadblocks, flooding, landslides, and heavy traffic
- Anti-spam duplicate rejection within 500 meters and 2 hours of existing reports
- Proactive arrival advice engine recommending umbrella or warm layers based on destination ETA

USER WORKFLOW

Resolving a Roadside Breakdown in Under 90 Seconds

An end-to-end trace of GuzoMate's real-time spatial retrieval and community safety pipeline.

STEP 1: VOICE QUERY

Traveler in Distress

"የመኪናዬ ሞተር ተበላሸ፣ በአቅራቢያዬ መካኒክ አለ?"

Driver taps voice button or speaks hands-free in Amharic. The frontend captures audio and extracts coordinates.

STEP 2: SPATIAL RAG

BGE-M3 + PostGIS

ST_DWithin($1, $2, 5000)

1,024-dim embedding model classifies repair intent. PostGIS executes geodesic radius query on addis_geo_points.

STEP 3: RECOMMENDATION

Verified Place Card

Grounded Place Output

Displays nearest verified automotive workshop, direct phone contact, verified services, and OSRM driving directions.

STEP 4: SAFETY ALERT

Community Incident Log

Geotagged Hazard Alert

Traveler logs the stranded vehicle. Approaching drivers receive real-time HUD alerts warning them of the obstruction.

AI & ML ARCHITECTURE

Dense Multilingual Embeddings with Zero Hallucination

Language parity across Latin, Ethiopic (Ge'ez), and Qubee orthographies.

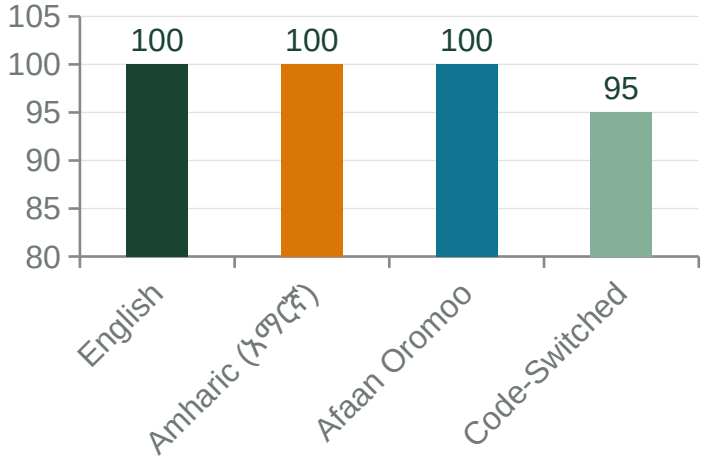
PIPELINE SPECIFICATIONS

Grounded Fact Retrieval

- **BAAI/bge-m3 Model:** Dense 1,024-dimensional embeddings capturing semantic parity across English, Amharic, and Afaan Oromo.
- **Supabase pgvector Index:** Cosine similarity search (<=>) indexed directly in PostgreSQL alongside spatial PostGIS coordinates.
- **Strict Evidence Guardrails:** Responses restricted to <EVIDENCE> tags. System is strictly instructed to disclose missing data rather than invent it.
- **Multi-Tier Query Planner:** Extracts language, intent flags (PLACES, WEATHER, ROUTING, INCIDENTS), and dynamic search radius in meters.

BENCHMARK VERIFICATION

Language Classification Benchmark



GEOSPATIAL FOUNDATION

18,833 Verified Locations Across Addis Ababa

Extracted from OpenStreetMap, deduplicated within 35m spatial radius, and hosted on Supabase PostGIS.

1,730 — **Cafes & Coffee Hubs**

Traditional Ethiopian coffee hubs, roasteries, and tea houses along major corridors.

1,459 — **Banks & Cash ATMs**

Commercial Bank of Ethiopia, private bank branches, cash points, and currency exchanges.

1,259 — **Hotels & Lodging**

Hotels, guest houses, and lodges categorized with verified addresses and coordinates.

421 — **Emergency Care & Fuel**

315 verified pharmacies and 106 hospitals & health clinics with 111 service stations.

POSTGIS OPTIMIZATION: ST_DWithin spherical index queries • Sub-50ms execution • Progressive radius expansion (3km local → 8km neighborhood → 18km city-wide).

SAFETY PROTOCOL

Crowdsourced Incidents with Democratic Verification

Preventing misinformation with strict duplicate detection and peer consensus.

01

Geotagged Incident Logging

Users submit road incidents (Accident, Roadblock, Flooding, Landslide) with GPS coordinates and timestamps.

500m / 2hr Anti-Spam Check

02

Peer Corroboration Voting

Nearby commuters receive verification prompts. Reaching ≥ 3 community confirmations transitions status from PENDING to ACTIVE.

3-Vote Activation Threshold

03

Automated Lifecycle Expiry

A background job runs every 5 minutes, automatically expiring stale incidents based on severity (2h traffic to 72h construction).

Auto-Expiry Background Worker

INCIDENT LIFECYCLE STATES IN GUZOMATE

- **Status: PENDING (Amber) —** Newly reported; awaiting 3 community confirmations.
- **Status: ACTIVE (Emerald) —** Confirmed hazard; active in radar queries.
- **Status: EXPIRED / RESOLVED —** Cleared by worker or moderator.

TRAVEL SERVICES

Proactive Arrival Advice & Multi-Modal Routing

Integrating live microclimate conditions, transit schedules, and OSRM route guidance.

WEATHER & MICROCLIMATE

Live Atmospheric Forecasting

- Integrated WeatherAPI and Open-Meteo fallback cache
- Hourly and daily precipitation mm, wind kph, and temperature
- WMO weather code descriptions localized in EN, AM, OM
- Elevation adjustments for highland microclimates

PROACTIVE ADVISORY

Intelligent Arrival Advice

- Evaluates destination weather matching arrival ETA
- Recommends umbrella when precipitation probability ≥ 35%
- Recommends warm layers when temperature ≤ 15°C
- Recommends sun protection when temperature ≥ 23°C

MULTI-MODAL ROUTING

Transit & OSRM Engine

- GTFS transit synthesis: 199 bus routes and 1,355 bus stops
- Addis Ababa Light Rail Transit (LRT) stop coordinates
- OSRM driving and walking routing with turn-by-turn maneuvers
- Localized route instructions in English, Amharic, and Afaan Oromo

SYSTEM ARCHITECTURE

Production-Grade Technology Stack

Strictly verified open-source components and cloud services implemented in the codebase.

CLOUD & DATABASE

AWS + Supabase

PostGIS + pgvector

- AWS Cloud hosting in eu-west-1 Ireland
- Supabase PostgreSQL 15 database
- PostGIS 3.3 for spatial geography queries
- pgvector indexing 1,024-dim BGE-M3 vectors
- Prisma ORM 6.19 for schema migrations

BACKEND API

Node.js v22 + Express

TypeScript REST API

- Modular domain services architecture
- Modules: Places, Assistant, Travel, Incidents
- Integrated WeatherAPI & Open-Meteo cache
- GTFS transit adapter & OSRM routing
- Winston structured production logging

AI & MULTILINGUAL

BGE-M3 + Gemini

Grounded RAG

- BAAI/bge-m3 dense 1,024-dim vector model
- Google Gemini 3.5 Flash Lite query planner
- LangChain RAG context boundary assembler
- Addis Voices 2 multilingual speech adapter
- Deterministic rule-based safety circuit breaker

FRONTEND & 3D

React 19 + Three.js

Highland Modern UX

- Vite build tool with zero TypeScript errors
- TailwindCSS custom Highland design tokens
- Three.js 3D wireframe terrain mesh canvas
- Leaflet interactive high-contrast map
- Framer Motion smooth micro-animations

MARKET IMPACT

Serving Real Traveler Segments in East Africa

Addressing high-friction transit scenarios across urban centers and regional highway corridors.

URBAN COMMUTERS

Daily City Transit Riders

Discover minibus routes, check destination weather arrival advice, and track verified road slowdowns during rush hours.

COMMERCIAL DRIVERS

Long-Haul Truckers & Freight

Identify verified heavy-duty repair garages, fuel stations, and road obstruction alerts along intercity freight corridors.

VISITORS & TOURISTS

Tourists & Eco-Adventurers

Overcome language barriers instantly; explore vetted cultural coffee spots, hotels, viewpoints, and emergency clinics.

REGIONAL TRAVELERS

Newcomers to the Capital

Ask queries hands-free by voice in native Amharic or Afaan Oromo; receive clear, grounded directions without digital literacy friction.

✦ GUZOMATE CLOSING SUMMARY ✦

Making Every Journey Safer and Easier.

An intelligent, multilingual co-pilot bridging local geography, dense vector RAG, and community safety across Ethiopia and beyond.

LIVE APPLICATION

Deployed Web Platform

Interactive React 19 UI with 3D WebGL Highland terrain mesh, Leaflet spatial map, and voice assistant.

➜ Live on AWS Cloud

SOURCE REPOSITORY

GitHub Monorepo

Complete codebase: TypeScript backend, React frontend, PostGIS migrations, and BGE-M3 RAG pipeline.

➜ github.com/eyob/GuzoMate-aws

DATABASE INFRASTRUCTURE

18,833 Verified POIs

Active Supabase PostgreSQL database in AWS eu-west-1 with PostGIS 3.3 and pgvector 1,024-dim index.

➜ Supabase Cloud Database